

Security in ZigBee networks



ZigBee

- ▶ Communication networks over IEEE 802.15.4
 - ♦ Low-rate wireless personal area network (LR-WPAN)
 - ♦ Low-power, low-cost, and low-data-rate wireless communication
 - ♦ Ideal for IoT and sensor networks
- ▶ IEEE 802.15.4 is used in other comm. stacks
 - ♦ For instance, 6LoWPAN

IEEE 802.15.4 technical features

- ▷ Low data rates
 - ♦ 20 kbps up to 250 kbps
- ▷ Range
 - ♦ 10–100 m
- ▷ Network Topologies
 - ♦ Star and peer-to-peer topologies
- ▷ MAC Access Method
 - ♦ CSMA/CA (Carrier Sense Multiple Access w/ Collision Avoidance)
 - ♦ Acknowledgement and retransmission
- ▷ Addressing
 - ♦ 16-bit short or 64-bit IEEE extended addresses
- ▷ Security
 - ♦ AES-128 encryption and integrity control

ZigBee node types

▷ Full-function device (FFD)

- ◆ Coordinator of a PAN
 - Talks with any other network node
- ◆ May relay messages in the network

▷ Reduced-function devices (RFD)

- ◆ Never a coordinator
- ◆ Interact with coordinators

Typical ZigBee networks

▷ Peer-to-peer

- ◆ Simple connection between two nodes

▷ Star

- ◆ An FFD acts as a PAN coordinator
- ◆ A set of nodes connect to the coordinator

▷ Meshed

- ◆ Variable topologies

Security

- ▷ Applied at link layer
 - ♦ But managed by upper layers
- ▷ Uses AES-128 for confidentiality and integrity control
 - ♦ Frame replay is tackled

Network key

- ▷ Network-wide key for encryption & decryption
 - ◆ All devices that are authorized to join the network have a copy of the key and use it to encrypt and decrypt all network messages
 - ◆ Used for secure broadcasting (group key)
 - ◆ Distributed by a Trust Center
- ▷ Has an associated sequence number
 - ◆ To identify a particular instance of the key
 - 0 to 255, round-robin
 - ◆ When the network key is updated, the sequence number is incremented
 - To allow devices to identify which instance of the network key has been used to secure the packet data

Trust Center

- ▶ May control the authentication in a secure network
 - ◆ All devices entering the network are temporarily joined to the network until it decides whether to allow its admission
 - ◆ The parent of the newly joined device acts as a relay between the Trust Center and the joining device
 - ◆ Only authentication messages can be sent to or from the device until it is fully joined and authenticated
- ▶ The Trust Center can:
 - ◆ Send a copy of the current network key to the joining device
 - ◆ Send the parent a command to remove it
 - ◆ Ignore the request
 - Parents will silently remove the device from the network if it does not receive a network key within 2 seconds.

Other keys

▷ Master key

- ◆ Secret shared between two nodes
- ◆ Shared out-of-band / manually
- ◆ Used to derive the link key

▷ Link key

- ◆ Used for unicast secure communication
- ◆ Unique between each pair of devices
- ◆ Derived from the Master Key using the SKKE (Symmetric-Key Key Establishment) protocol